

Sample BIS Product Standard B — Electrical Product Safety Requirements

Synthetic dataset for BIS Navigator RAG testing. Not an official BIS document.

Clause 1 · Scope

- 1.1 This synthetic standard specifies safety requirements for Product B used in controlled electrical applications.
- 1.2 It covers construction, insulation, electrical performance, testing and marking.
- 1.3 The requirements are intended to support consistent evaluation of product safety.

Clause 2 · Product Characteristics

- 2.1 Product B shall have characteristics appropriate to its declared operating conditions.
- 2.2 Rated values shall be stated in the technical specification.
- 2.3 Safety-related characteristics shall be verified by the applicable test procedure.
- 2.4 Components shall be installed in accordance with the documented design.

Clause 3 · Electrical Safety

- 3.1 Insulation shall provide adequate protection under the declared operating conditions.
- 3.2 Electrical connections shall be secure and suitable for the intended current.
- 3.3 Accessible conductive parts shall satisfy the applicable protection requirement.
- 3.4 The product shall not present an unacceptable electrical hazard during normal operation.

Clause 4 · Mechanical Safety

- 4.1 Enclosures shall provide suitable mechanical protection.
- 4.2 Moving or removable components shall be secured against unintended displacement.
- 4.3 Mechanical strength shall be assessed using the prescribed test conditions.

Clause 5 · Performance Tests

- 5.1 The product shall undergo the applicable performance tests.
- 5.2 Tests shall be conducted using calibrated or otherwise suitable equipment.
- 5.3 Test conditions shall be recorded with the test result.
- 5.4 Repeated tests shall use the same specified procedure unless an approved deviation is documented.
- 5.5 Failed tests shall be investigated and documented before conformity is concluded.

Clause 6 · Routine Inspection

- 6.1 Routine inspection shall be performed at the defined manufacturing stage.
- 6.2 Inspection shall include visual examination and applicable functional checks.
- 6.3 Inspection results shall be traceable to the product or batch.
- 6.4 Defective units shall be segregated and controlled.

Clause 7 · Marking and Labels

- 7.1 Product B shall display the required identification information.
- 7.2 Rated values and safety-related labels shall be legible.
- 7.3 Marking durability shall be verified where required by the conformity procedure.

Clause 8 · Test Reports

8.1 Test reports shall identify the product, sample, procedure and result.

8.2 Reports shall include relevant measurement values and acceptance criteria.

8.3 Supporting records shall be retained for verification.

Clause 9 · Certification Route

9.1 The applicable certification route shall be determined from the product category.

9.2 The applicant shall submit the documents specified by the applicable route.

9.3 Additional evaluation may be required when safety-critical characteristics are changed.